



Assignment (1)

Student name : **Student Number:**.....

Exercises :

1. Write a Dart program that prints the following information:

- Your name
- Your age
- Your major

Each item should be printed on a separate line.

2. Write a Dart program that calculates and prints the following for two numbers:

- Sum
- Multiplication
- Subtraction
- Division
- Remainder

3. Write a Dart program to evaluate and print the results of the following expressions:

- a. $-5 + 8 * 6$
- b. $(55 + 9) \% 9$
- c. $20 + -3 * 5 / 8$
- d. $5 + 15 / 3 * 2 - 8 \% 3$

4. Write a Dart program to print your complete personal information.
Each item should be printed on a new line:

- Name
- University
- University Number
- College
- Address
- Marital Status
- 5. Write a Dart program to print the following shape:

5. Write a Dart program to print the following shape:

```
+ " " " " " " " " +
```

```
[ | o o | ]
```

```
| ^ |
```

```
| ' ' |
```

```
+-----+
```

6. Write a Dart program that calculates the area of a rectangle using the following formula:

Area = length × width

Hint:

Read the length and width from the user.

7. Write a Dart program that calculates the area of a circle using the following formula:

Area = $\pi \times r^2$

($\pi = 3.14$)

Hint:

Read the radius from the user.

8. Write a Dart program to convert an amount in shekels to dollars.

Hint:

- Read the amount in shekels from the user
- (1 Dollar = 3.25 Shekels)

9. Write a Dart program that converts temperature from Fahrenheit to Celsius using the following formula:

$$\text{Celsius} = 5 \times (\text{Fahrenheit} - 32) / 9$$

Hint:

Read the temperature in Fahrenheit from the user.

10. Write a Dart program that reads a value in inches and converts it to meters.

Hint:

- Read the value in inches from the user
- (1 inch = 0.0254 meter)

11. Write a Dart program that reads an integer between 0 and 1000 and calculates the sum of its digits.

Hint:

Read the number from the user.

12. Write a Dart program to convert a number of minutes into years and days.

Hint:

Read the number of minutes from the user.

Test Data:

Input: 3456789

Expected Output:

3456789 minutes is approximately 6 years and 210 days

Prepared by:

Eng. Mohammed Iyad Al-Laham